

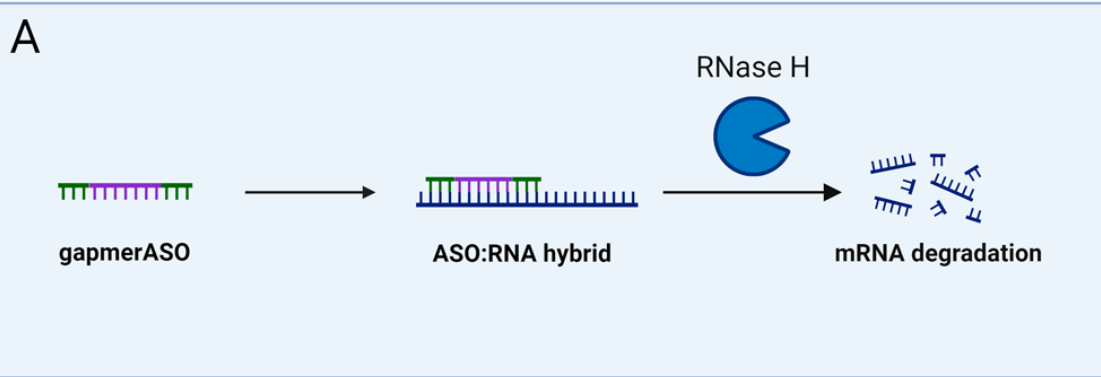
- Administration of **genetic material (DNA or RNA)** to modify or manipulate the expression of a gene product or to alter the biological properties of living cells for **therapeutic use** (FDA)
 1. Gene replacement : the delivery of a functional gene to replace a non working gene
 2. Gene silencing : Inactivation of a mutated gene that has become toxic to cells
 3. Gene addition: overexpression of a “foreign” or exogenous gene to impact cellular function
 4. Gene editing: a permanent manipulation of a gene in a patient’s genome

GENE THERAPY DELIVERY METHODS

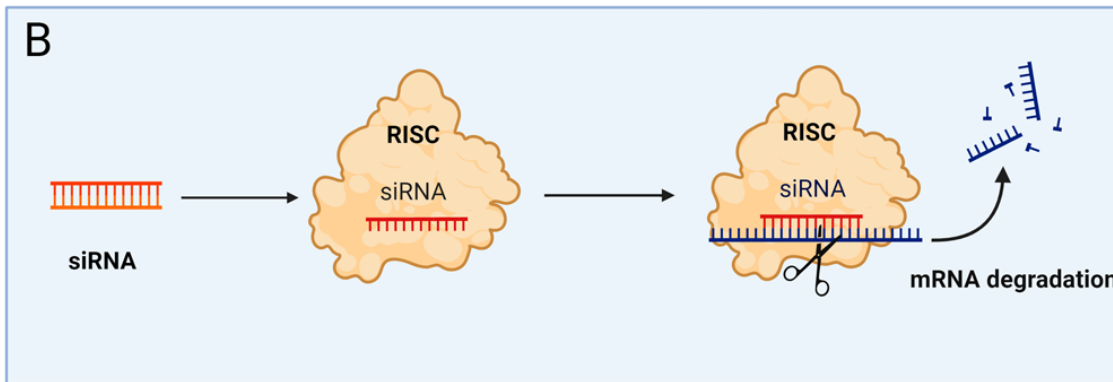


Virus Family	Transgene size	Is DNA integrated into host genome?	Duration of gene expression	Tropism	Immunogenicity
Adenovirus	8-38 kb	No	Short	Dividing and nondividing cells	Typically immunogenic and pre-existing immunity
AAV	4.5 kb	No	Long	Broad tropism, dividing and nondividing cells	Typically low, though some pre-existing
Retrovirus	8-9 kb	Yes	Short	Broad tropism	Low immunogenicity
Lentivirus	9 kb	Yes	Long	Dividing and nondividing cells	Low Immunogenicity
Herpes	~50 kb	No	Long in CNS, short elsewhere	Nondividing and dividing cells, infects CNS	High levels of pre-existing immunity
Rhabdovirus	4.5 kb	No	Short	Broad tropism	Immunogenic
Vaccinia	30 kb	No	Short	Broad tropism	Immunogenic

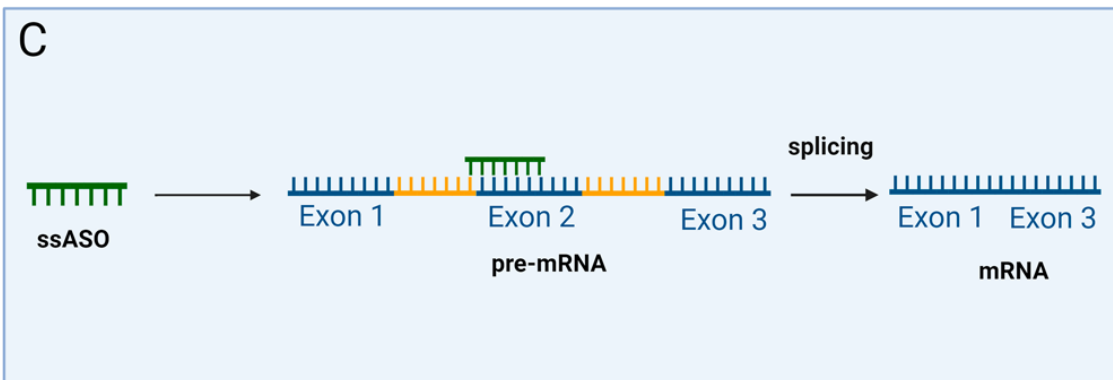
GENE SILENCING



Gapmer ASOs bind to their target mRNA to form DNA:RNA hybrids that recruit **RNase H** for the cleavage of mRNA



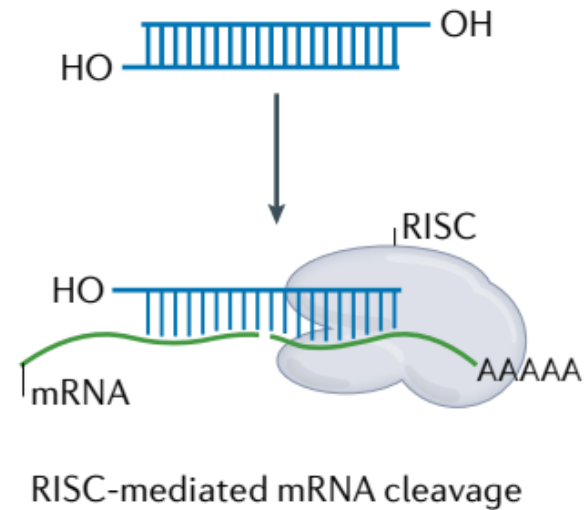
siRNAs enter the cell as double strands and are incorporated into the **RNA-induced silencing complex (RISC)**, where the siRNA gets unwound. The siRNA can now bind to its target mRNA and induce cleavage



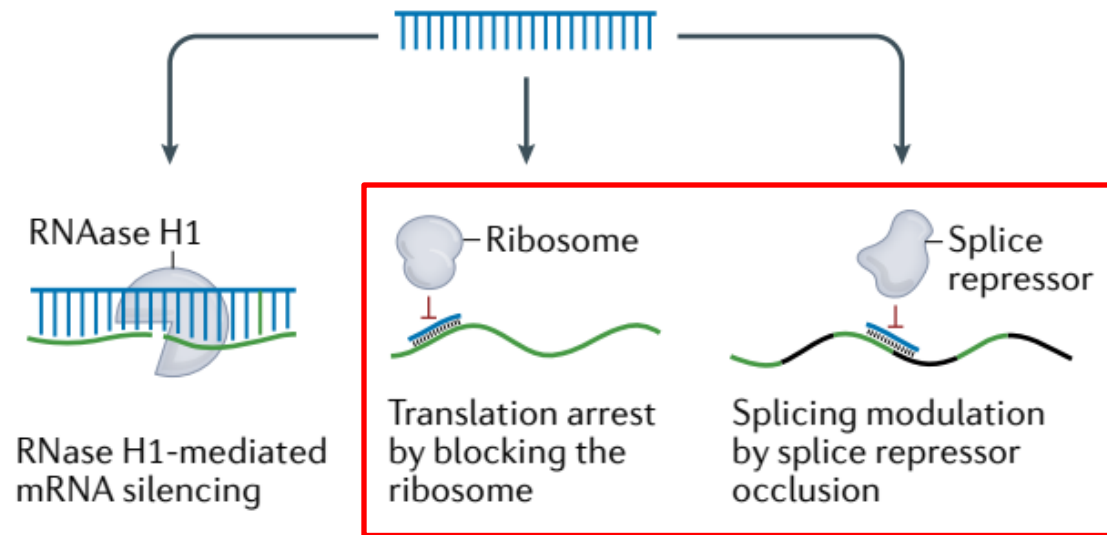
Splice-switching ASOs (ssASOs) bind to pre-mRNA and modulate the splicing process.

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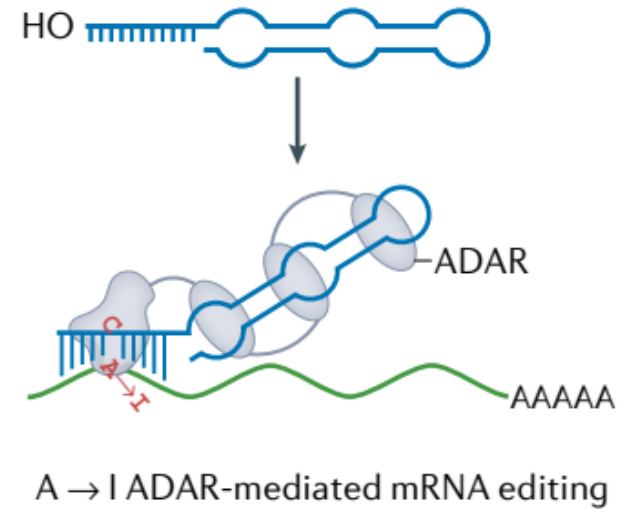
siRNA: ~13 kDa



ASO: ~6 kDa

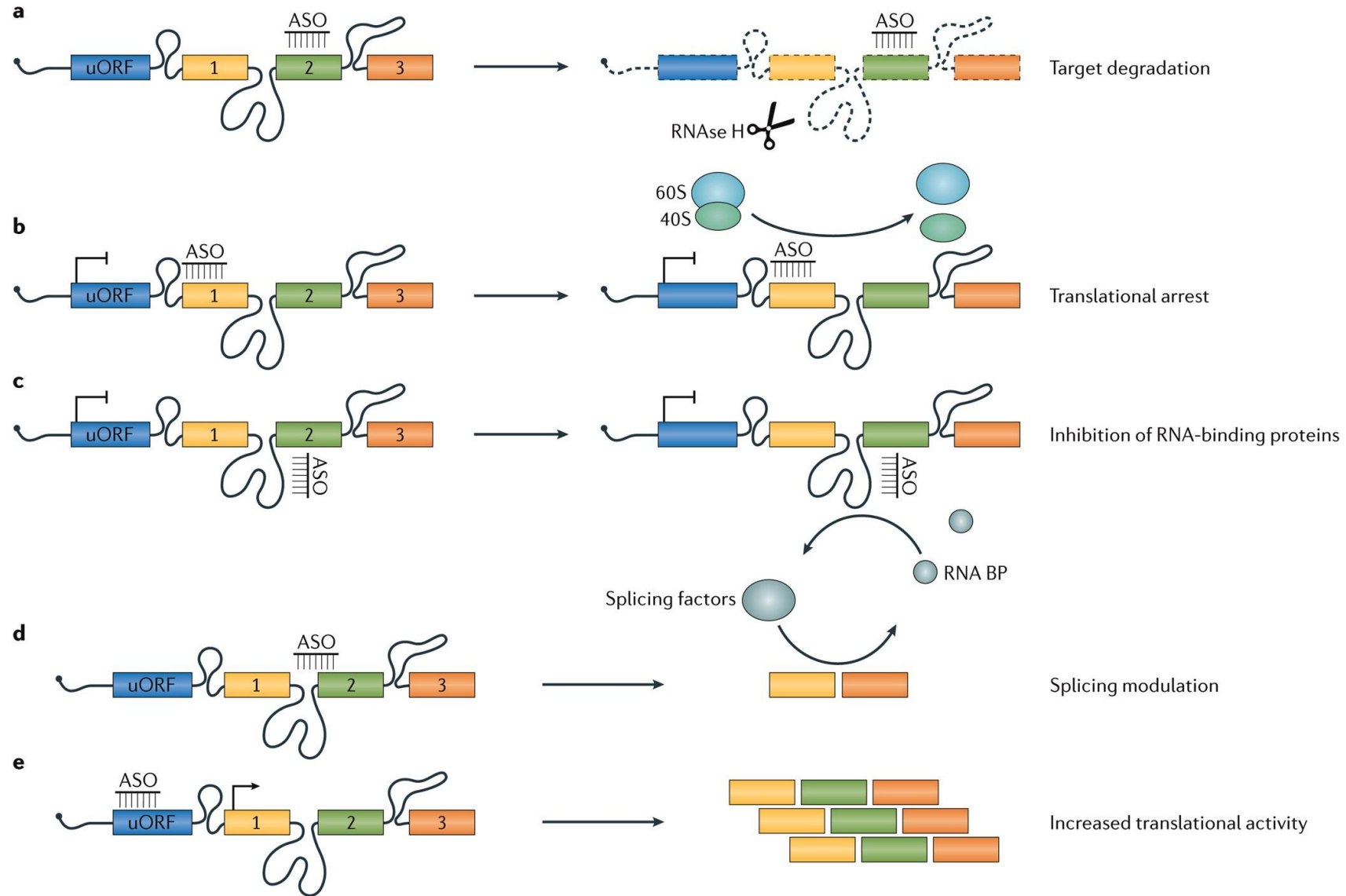


ADAR-oligonucleotides: ~24 kDa

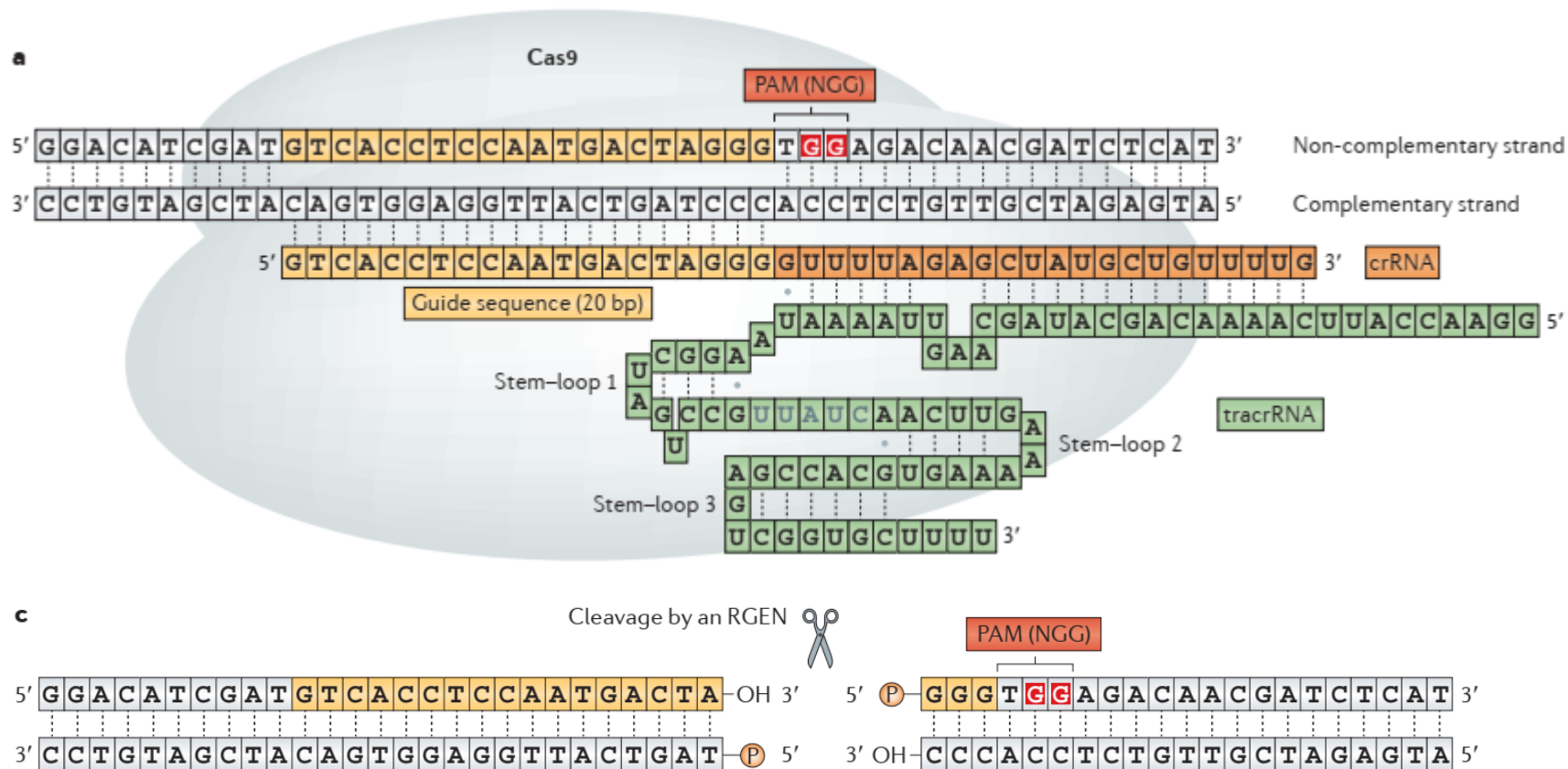
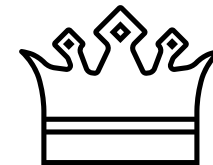


Nuclear delivery (pre-mRNA)

ASO-MEDIATED GENE REGULATION



RNA-guided engineered nuclease (RGENs)



crRNA: CRISPR RNA, 표적 DNA와 상보결합하는 **20bp의 guide sequence** (protospacer)를 가지고 있음.

tracrRNA : trans-activating crRNA, 표적과는 상관없이 동일한 염기서열

crRNA + tracrRNA : single guide RNA (sgRNA)

PAM : Protospacer adjacent motif (표적 DNA에 존재)

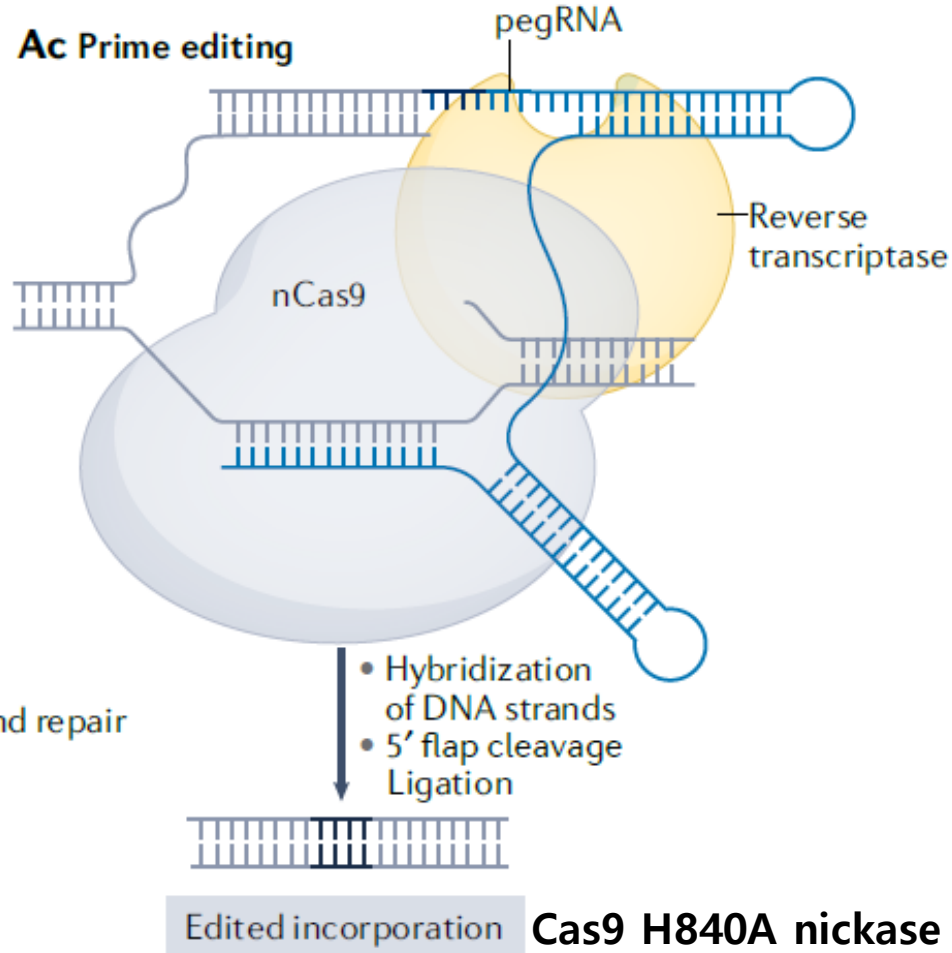
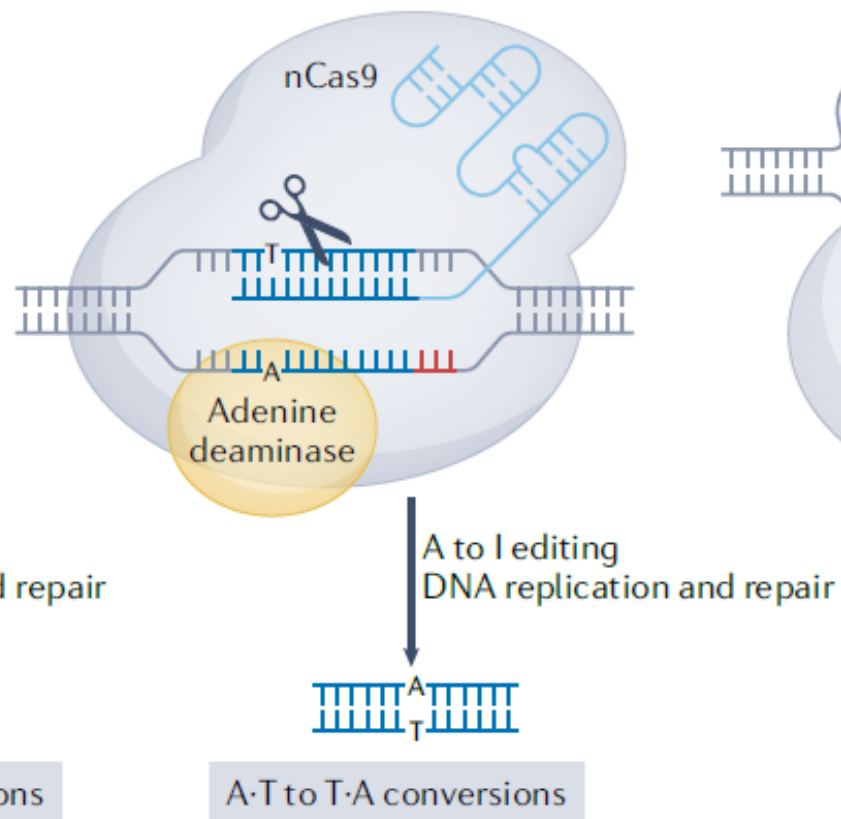
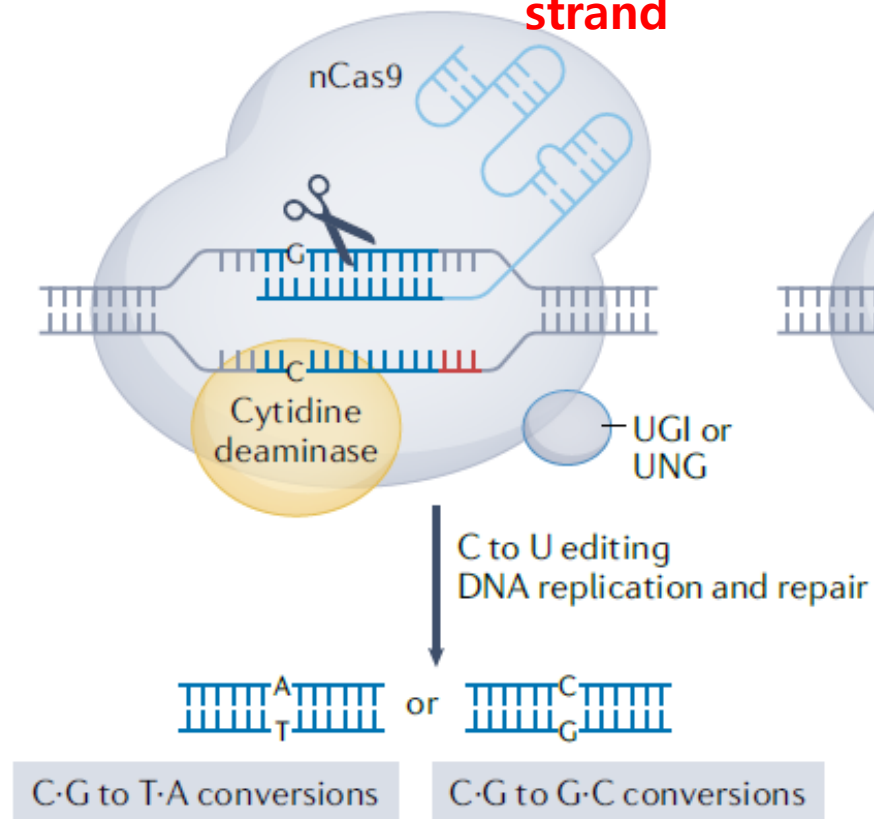
Next-generation genome editors



Cas9 D10A nickase

-> Inactive RuvC domain ->
do not cleave non**target**
strand

Ab Single base editing



Cas9 H840A nickase

-> Inactive HNH domain
-> do not cleave **target**
strand

Do not induce **Double strand breaks!**